Design and implementation of teacher AI literacy assessment system

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Abstract

Conducting teacher AI literacy assessments is a key task in implementing the digital transformation of education. In 2024, UNESCO released the AI Competency Framework for Teachers, providing systematic guidance for developing AI literacy for teachers worldwide. This article first systematically introduces the framework's five aspects and three progression levels. Based on this framework, it designs and develops a teacher AI literacy assessment system, aiming to provide tools to support the diagnosis, training, and personalized development of teachers' AI competences.

CCS Concepts

• **Information systems** → World Wide Web; Web applications.

Keywords

Teacher AI literacy, assessment system, AI-CFT, digital transformation, teacher development

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1 Introduction

Artificial intelligence (AI), as an important technological innovation force in the field of education, has been deeply involved in teaching management, learning resource optimization, personalized learning and other aspects [1]. The widespread application of AI is prompting the traditional teacher-student relationship to transform into a "teacher-AI-student" triangular interactive model, which puts forward new requirements for the role positioning and professional quality of teachers. Teachers must not only have the ability to evaluate and effectively use AI tools, but also be able to guide students to make reasonable use of AI, promote their personalized learning and comprehensive development, and thus improve the overall teaching effect [2].

However, the current global improvement of teachers' AI literacy still faces many challenges. According to statistics, as of 2022, only seven countries have formulated teacher AI competency frameworks or related professional development plans. Teachers in most countries lack scientific and systematic AI literacy development guidance, making it difficult to fully utilize AI technology to serve teaching practice [3]. This also highlights the urgency and necessity of building a scientific and systematic teacher AI literacy assessment system.

In 2024, UNESCO released the AI Competency Framework for Teachers (AI-CFT), which systematically describes and defines teachers' AI competences from five aspects and three progression levels, and constructs a corresponding evaluation matrix. This framework provides a theoretical foundation and practical tools for self-diagnosis and capacity building of teachers' AI literacy, as well as for educational institutions to conduct training and formulate policies.

Based on the UNESCO AI-CFT framework, this paper designs and develops an online teacher AI literacy assessment system. This system aims to survey teachers' current AI literacy status, encompassing the five aspects and three progression levels defined by the AI-CFT framework, and comprehensively assessing teachers' AI knowledge, skills, and attitudes. Through systematic assessment and data analysis of teachers' AI literacy, this paper aims to identify key current issues and challenges, propose targeted improvement recommendations, and provide a theoretical basis and practical path forward for promoting digital transformation in education and teacher professional development.

2 UNESCO AI Competency Framework for Teachers(AI-CFT Framework)

2.1 Teacher AI literacy

What is AI literacy? Long et al. proposed "AI literacy as a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace"[4]. Teacher AI literacy refers to teachers' understanding of the basic principles of AI technology in an educational context, their ability to evaluate the applicability of AI tools, and their ability to creatively and ethically use AI technology in the teaching process[5]. Ng et al. further suggested a set of teacher's AI competencies, including "using basic applications, managing information, creating learning content, and connecting their students via technology"[6]. Teacher AI literacy not only includes the basic cognition and operational competences of artificial intelligence technology, but also covers the ability to

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Table 1: UNESCO AI-CFT Framework.

Aspects	Progression		
	Acquire	Deepen	Create
Human-centred mindset	Human agency	Human accountability	Social responsibility
Ethics of AI	Ethical principles	Safe and responsible use	co-creating ethical rules
AI foundations and applications	Basic AI techniques and applications	Application skills	Creating with AI
AI pedagogy	AI-assisted teaching	AI-pedagogy integration	AI-enhanced pedagogical transformation
AI for professional development	AI enabling lifelong professional learning	AI to enhance organizational learning	AI to support professional transformation

innovatively apply AI in teaching practice to promote teaching effectiveness and enhance students' learning experience. In addition, teachers must also be sensitive to issues related to AI ethics, data security and privacy protection, and be able to reasonably guide students' interactions with AI while protecting students' rights and interests. Improving teachers' AI literacy will help them better adapt to the digital transformation of education and leverage the advantages of AI tools in curriculum design, personalized teaching, and academic assessment, thereby promoting educational equity and high-quality development. Currently, international organizations and national education departments attach great importance to cultivating teachers' AI literacy and have successively launched relevant competency frameworks and training programs.

2.2 AI-CFT Framework

In 2024, UNESCO released the AI Competency Framework for Teachers (AI-CFT), aiming to provide a systematic guide for developing AI literacy for teachers worldwide [7]. The framework describes the AI literacy that teachers should possess across five aspects and three progression levels: Human-Centred Mindset, Ethics of AI, AI Foundations and Applications, AI Pedagogy, and AI for Professional Development. The levels are Acquire, Deepen, and Create (see Table 1).

2.2.1 Five Aspects of AI-CFT Framework. The Human-centered Mindset aspect clarifies the values and attitudes teachers should uphold when promoting human-machine collaboration. Its core emphasis is that AI should be viewed as a tool to assist education, not as the ultimate goal of education. Teachers should scientifically evaluate and rationally select the applicability of AI tools based on the perspective of promoting students' holistic development and individual growth, ensuring that technology consistently serves the needs of human development.

The Ethics of AI aspect stipulates the basic ethical principles, relevant laws and regulations, and best practices that teachers should understand, apply, and guide their practice in using AI. It focuses on potential issues raised by AI in areas such as privacy protection, fairness, explainability, and algorithmic bias. Teachers should develop a critical awareness of the ethical risks of AI applications in education, preventing them from negatively impacting educational equity and student rights.

The AI Foundations and Applications aspect requires teachers to master core AI concepts and transferable skills, enabling them to scientifically select, apply, and creatively customize AI tools within student-centered, AI-assisted teaching environments. This includes understanding the basic working principles of AI (such as algorithms and machine learning), familiarizing themselves with the operation of various AI tools, and possessing the ability to evaluate and critically apply them.

The AI pedagogy aspect addresses the skills teachers need to purposefully and effectively integrate AI into their teaching practices. This includes the ability to select and validate appropriate AI tools and apply them to curriculum design, classroom instruction, student learning, socialization, student care, and learning assessment. Teachers should be adept at employing innovative teaching methods, such as personalized learning and virtual simulations, to enhance the scientific and innovative nature of their teaching.

The AI for professional development aspect emphasizes that teachers should be able to leverage AI to promote lifelong professional learning, support collaborative professional development, and explore career transitions. Through AI technology, teachers can optimize teaching reflection, enhance the effectiveness of academic communication, and achieve personalized career development paths, thereby continuously adapting to the changing and challenging educational environment [8].

2.2.2 Three Progression Levels of AI-CFT Framework. The progression levels of in the AI CFT helps assess teachers' existing AI competences and clarify their professional learning goals. The "Acquire" level is for teachers who lack AI-related knowledge and skills. Training courses at this level aim to support teachers in mastering the basic AI competences and literacy required for teaching in various settings. The "Deepen" level is for teachers who already possess some AI knowledge and experience applying AI in educational practice. Training courses at this level aim to support teachers in becoming core teachers capable of fully utilizing AI. They demonstrate a human-centered perspective and ethical behavior in their analysis and decision-making processes, have a deep understanding of AI concepts, and can effectively apply AI to teaching activities and professional learning [9]. The "Create" level is for teachers with solid AI knowledge and skills and extensive experience in applying AI in education. Training courses at this level aim to cultivate teachers with sound AI knowledge and competences, enabling them to become not only experts in education but also drivers of AI-based

Table 2: Understanding of Three Progression Levels

levels	Acquire	Deepen	Create
Description	You will be able to understand the basic concepts of AI, know how to select and use common AI tools, and begin to think about the potential risks and benefits of AI in teaching.	You are proficient in using AI tools to design teaching strategies, understand the ethical issues of AI, and ensure the safe and responsible use of AI in teaching.	You can use AI tools innovatively to solve complex teaching problems, and even participate in the development or improvement of AI tools to promote educational change.
Applicable scenarios	You are new to AI, learning how to integrate AI tools into everyday teaching, and are paying attention to AI's impact on students and instruction.	You have already mastered several AI tools and can integrate AI with teaching methods to design more effective learning activities, while paying attention to AI's impact on student learning and the teacher's role.	You have a deep understanding of AI, can customize or develop AI tools to meet instructional needs, and drive innovative AI applications within schools or education systems.

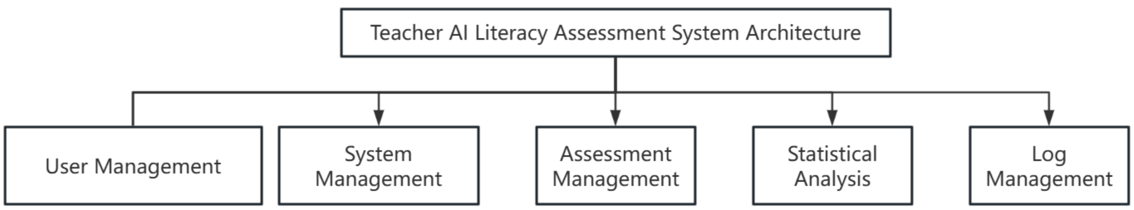

Figure 1: Functional Module Diagram of the Teacher AI Literacy Assessment.

educational change. These teachers are able to innovatively apply AI to education and collaborate with the community to explore how AI can achieve desired transformations in teaching practice. Table 2 shows the understanding and applicable scenarios for each level.

This article uses this framework as a theoretical model to design and develop a teacher AI literacy assessment system. This system aims to systematically and meticulously measure and analyze teachers' AI literacy using a scientific indicator system and hierarchical standards. By combining five core aspects with three progression levels, the resulting assessment system not only helps diagnose the current status and shortcomings of teachers' AI literacy but also provides data support for subsequent professional development and training path planning.

3 Design of a Teacher AI Literacy Assessment System

3.1 System Design

To facilitate the assessment of teachers' AI literacy, a modular design approach is required. The system is divided into five modules: user management, system management, assessment management, statistical analysis management, and log management (see Figure 1 for details). The functions of each module are as follows:

User Management: Adding users, modifying users, deleting users, and changing passwords.

System Management: Setting system parameters, timeout management, user permission management, and system UI settings.

Assessment Management: Adding assessments, managing assessment scales, managing assessment time, and managing assessment data.

Statistical Analysis Management: Managing statistical aspects, charts, and indicators.

Log Management: Viewing, searching, and exporting logs.

3.2 Assessment Process

The complete assessment process consists of four steps: scale design, scale distribution, scale completion, and scale evaluation (see Figure 2 for details). The assessment steps are as follows:

Scale Design: This step designs the scale based on the AI Literacy Framework and is divided into five sections: Human-centered Mindset, Ethics of AI, AI Foundations and Applications, AI Pedagogy, and AI for Professional Development.

Scale Distribution: After the scale design is completed, it will be notified to the teachers to be assessed via SMS, WeChat, or other messaging systems. Teachers click the URL in the message and, after verification, will be able to enter the scale assessment.

Scale Completion: Teachers select the options that best suit their individual circumstances. After completing each assessment, they submit the scale.

Scale Evaluation: The system compiles statistics based on the scale data submitted by the teacher, assesses the teacher's AI literacy, and generates an assessment report.

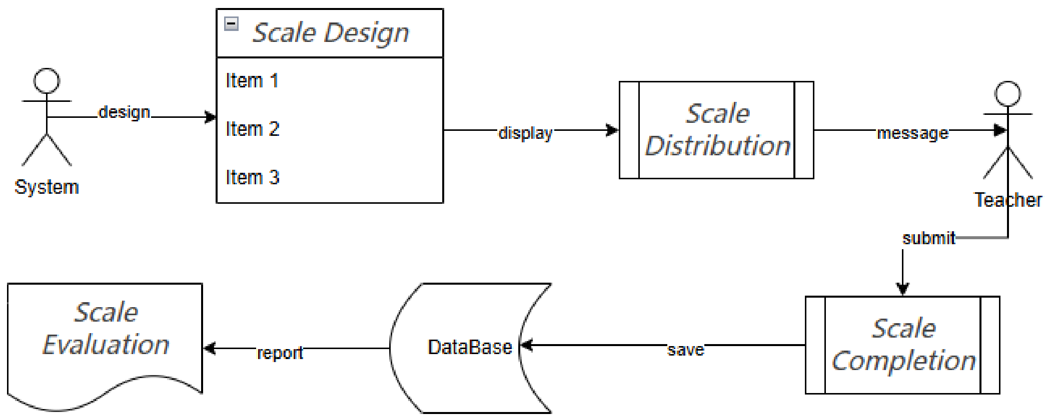

Figure 2: Flowchart of Teacher AI Literacy Assessment.

4 Implementation of the AI Teacher Literacy Assessment System

To reduce development costs, the system utilizes an open-source software architecture, with an operating system of Ubuntu 24.0, basic software of SurveyKing, a database of MySQL, and a deployment method of Docker-Compose. Surveyking is an open-source survey and statistics software available free to everyone. It includes multiple functional modules, including online exams, questionnaires, public queries, a question bank, 360-degree assessments, and a voting system. It supports multiple languages and is currently the most widely used free survey and questionnaire system. It can be deployed in various ways, including source code installation and running in a Docker container. The simplest and fastest way is to install it in a Docker container [11].

4.1 Installing Docker and Docker-Compose

Docker is the most prominent example of PaaS. Based on virtualization technology, it packages the software runtime environment into a single file, enabling developers to quickly deploy information systems [10]. It is currently the mainstream software system deployment method in the industry [12].

```
#Update the system to the latest stable version
sudo apt update
#Install Docker and Docker Compose
sudo apt install docker.io
#Check the status of Docker and Docker Compose
sudo docker version
sudo docker compose version
```

4.2 Installing SurveyKing

There are many ways to install SurveyKing. This project uses Docker to minimize the workload. First, create the paths for SurveyKing and the database, then download the program configuration file and SQL file from the Internet. Finally, switch to the SurveyKing working directory and run the Docker packager. The Docker container will automatically build the relevant program packages. Once the corresponding packaging is completed, you can run the SurveyKing program.

```
#Create the program execution directory (program and database directories)
sudo mkdir -p ./surveyking/mysql
sudo mkdir ./surveyking/sqlshappywin11
sudo cd ./surveyking
#Download the database script file
#Download the Docker Compose file
sudo curl -o sqls/init-mysql.sql https://surveyking.cn/surveyking-docker/sqls/init-mysql.sql
sudo curl -o docker-compose.yml https://surveyking.cn/surveyking-docker/docker-compose.yml.example
#Run the SurveyKing program
docker-compose up –d
```

4.3 Creating and evaluating scales

Once the assessment scale is created and published, it can be displayed simultaneously on both PC and mobile devices, ensuring consistent assessment content regardless of device. After completing the assessment, participating teachers can instantly view their scores and understand their current digital literacy level (as shown in Figure 3). After the teacher completes the submission of the scale, the teacher's measurement data and current level will be automatically presented to the teacher based on the scale data submitted by the teacher and the scale measurement standards set by the system.

5 Application of Teacher AI Literacy Evaluation System

The teacher AI literacy evaluation system is based on the AI-CFT Framework. Through 30 questionnaire scales and collected teacher questionnaire data, it can provide real-time information on teachers' current AI literacy status, making it easier for teachers to conduct self-assessments. In addition, various teaching structures can publish questionnaire scales with their own characteristics based on

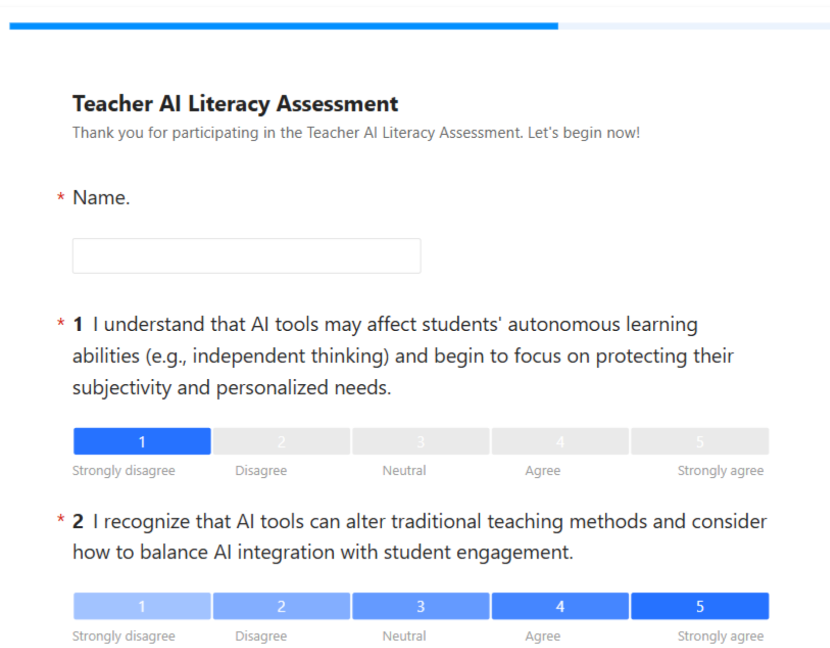

Figure 3: Creating a teacher AI literacy assessment scale.

the specific circumstances of the institution to collect AI literacy data of the institution's teachers. Based on this data, the system can conduct an overall evaluation of the institution's teachers' AI literacy and propose targeted improvement measures. The teacher AI literacy evaluation system solves the difficult problems of evaluating individual teachers' AI literacy and evaluating the AI literacy of teaching institutions from a system level.

6 Conclusion

The project used open-source software to rapidly build a teacher AI literacy assessment system. The system provides powerful competences such as scale design, collection, and analysis. Teachers can use the platform to quickly and easily assess their AI literacy, identifying weaknesses in digital literacy and facilitating targeted learning and improvement in their studies and work. The system can meet the needs of various educational institutions for quantitative assessment of teacher AI literacy and provide valuable insights for AI literacy assessment in the education sector.

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